
■ Enterprise Machine Learning MCQs (Detailed, No Answers)

Model Deployment & Infrastructure

1. In enterprise ML, which deployment strategy minimizes downtime during model updates?
 - A. Blue-Green Deployment
 - B. Shadow Deployment
 - C. Canary Deployment
 - D. Rolling Restart
 2. Which enterprise tool is most commonly used for orchestrating ML workflows across distributed systems?
 - A. Apache Airflow
 - B. Jenkins
 - C. GitHub Actions
 - D. Terraform
 3. Which approach ensures **version control** of ML models in production?
 - A. Model Registry
 - B. Feature Store
 - C. Data Lake
 - D. CI/CD Pipeline
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Feature Engineering & Data Handling

4. Which technique is best suited for handling **imbalanced datasets** in enterprise fraud detection systems?
 - A. Random Oversampling
 - B. SMOTE (Synthetic Minority Oversampling Technique)
 - C. Class Weight Adjustment
 - D. All of the above
5. In enterprise ML, which method is most effective for reducing **dimensionality** in high-volume datasets?
 - A. PCA (Principal Component Analysis)
 - B. One-Hot Encoding
 - C. Min-Max Scaling

- D. Logistic Regression
6. Which enterprise practice ensures **data lineage tracking** for ML pipelines?
- A. Metadata Management
 - B. Hyperparameter Tuning
 - C. Model Ensembling
 - D. Batch Normalization
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Model Monitoring & Governance

7. Which metric is most relevant for detecting **data drift** in production ML systems?
- A. KL Divergence
 - B. RMSE
 - C. F1 Score
 - D. ROC-AUC
8. In enterprise ML governance, which framework is often used to ensure **ethical AI practices**?
- A. ISO 27001
 - B. NIST AI Risk Management Framework
 - C. COBIT
 - D. ITIL
9. Which monitoring approach is best for detecting **real-time anomalies** in ML predictions?
- A. Batch Monitoring
 - B. Streaming Analytics
 - C. Offline Evaluation
 - D. Cross-Validation
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Scalability & Performance

10. Which distributed ML training technique reduces **communication overhead** across nodes?
- A. Data Parallelism
 - B. Model Parallelism
 - C. Federated Learning
 - D. Gradient Compression
11. Which enterprise ML framework is optimized for **large-scale GPU training**?
- A. Scikit-Learn
 - B. TensorFlow
 - C. PyTorch Distributed
 - D. XGBoost
12. In enterprise ML, which caching strategy improves **real-time inference speed**?

- A. Redis In-Memory Cache
 - B. SQL Indexing
 - C. Hadoop HDFS
 - D. Kafka Queues
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Bias, Fairness & Compliance

13. Which fairness metric evaluates whether different demographic groups receive similar positive outcomes?
- A. Equal Opportunity
 - B. Demographic Parity
 - C. Predictive Parity
 - D. Calibration
14. Which enterprise practice ensures compliance with **GDPR's "Right to Explanation"** in ML models?
- A. SHAP (SHapley Additive Explanations)
 - B. LIME (Local Interpretable Model-agnostic Explanations)
 - C. Counterfactual Explanations
 - D. All of the above
15. Which bias mitigation technique is most effective during **pre-processing**?
- A. Re-weighting samples
 - B. Adversarial debiasing
 - C. Fairness constraints in training
 - D. Post-hoc calibration
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Cloud ML & Enterprise Services

16. Which cloud service provides **end-to-end ML lifecycle management** including training, deployment, and monitoring?
- A. AWS SageMaker
 - B. Azure ML Studio
 - C. Google Vertex AI
 - D. All of the above
17. Which enterprise ML service supports **federated learning** across distributed devices?
- A. TensorFlow Federated
 - B. PyTorch Lightning
 - C. MLflow
 - D. Kubeflow
18. Which cloud ML practice ensures **cost optimization** during large-scale training?
- A. Spot Instances

- B. Auto-Scaling Clusters
 - C. Hybrid Cloud Deployment
 - D. All of the above
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Security & Risk Management

19. Which enterprise ML security risk involves attackers manipulating training data to degrade model performance?

- A. Model Inversion Attack
- B. Data Poisoning Attack
- C. Adversarial Example Attack
- D. Membership Inference Attack

20. Which enterprise ML security measure prevents **unauthorized model access**?

- A. Role-Based Access Control (RBAC)
- B. Encryption at Rest
- C. API Gateway Authentication
- D. All of the above